Melvin Lopez

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Education

University of Nevada, Las Vegas — Las Vegas, NV

B.S. in Electrical Engineering, Minor in Computer Science — *Expected May 2028*

- **GPA:** 3.4 / 4.0
- **Relevant Coursework:** Computer Organization & Design, Circuits (DC/AC Analysis), Electronics I, Signals & Systems, Digital Systems Design, Electromagnetics

Experience

Electrical Engineering & Computer Science Teaching Assistant — Aug. 2025 – Present

University of Nevada, Las Vegas — *Las Vegas, NV*

- Graded 50+ submissions weekly for CS 218 (Systems Programming) and EE 221 (Circuits II), writing targeted technical feedback that reduced repeated errors on later assignments
- Led weekly sessions guiding students through number systems, digital logic, and AC circuit analysis
- Debugged student code and circuit solutions one-on-one in office hours to resolve technical roadblocks quickly

Team Associate — June 2026 – Present

Walmart — *Las Vegas, NV*

- Stocked the detergent and liquor departments and answered customer assistance calls in a high-traffic store

Team Member — Mar. 2025 – June 2026

Steak 'n Shake — *Las Vegas, NV*

- Trained 5 new employees on operating procedures and managed order flow during 200+ order shifts over 15 months

Projects

Machine Learning Noise Cancellation | *Python, PyTorch, Real-Time Audio*

- Built a real-time noise suppression pipeline around a custom-trained neural network with under 50ms latency
- Trained an RNN on 1,000+ audio samples to predict STFT masks, outperforming an adaptive-filtering baseline
- Designed a dual audio-stream architecture with a thread-safe queue, eliminating device conflicts and overflows
- Debugged low-level Windows audio driver issues across 3+ hardware configurations for stable capture and playback

Digital Systems FPGA Slot Machine | *Verilog, FPGA (DE2-115), Hardware I/O*

- Designed a synchronous state machine with 6+ states in Verilog and demoed a fully functional device to instructors
- Isolated and corrected hardware faults across 10+ debugging iterations using waveform analysis
- Split design tasks with a project partner to deliver a working system on a fixed deadline

Leadership

Treasurer — Oct. 2026 – Present

IEEE, UNLV — *Las Vegas, NV*

- Manage chapter budget and track expenses for events and member programs
- Coordinate funding requests and reimbursements with university student organization offices

Technical Officer — Sept. 2026 – Present

Software Engineering Club, UNLV — *Las Vegas, NV*

- Lead club projects from proposal to execution, pitching ideas to members and shaping technical workshop topics
- Developing a large-scale team project for the RebelHacks hackathon

Technical Skills

Programming: C++, C, Assembly, Python, Verilog
Machine Learning: PyTorch, Neural Network Design & Training, Real-Time Signal Processing
Hardware: FPGA (DE2-115), Arduino, Circuit Analysis, Digital Logic Design
Software Tools: Git/GitHub, Linux, Intel Quartus, LTSpice, MATLAB, KiCAD